

THEORY OF COMPUTATION (CS302)

PDA Constructions

Learning Objectives

Understand Pushdown Automata (PDA), stack operations, acceptance methods, PDA design techniques, and CFG to PDA conversion.

Introduction to Pushdown Automata

A Pushdown Automaton (PDA) is a computational model that extends finite automata by adding a stack. PDAs are used to recognize context-free languages.

Why PDA is Needed

Finite Automata cannot recognize languages requiring memory. The stack in a PDA provides memory, allowing it to recognize context-free languages such as $a^n b^n$.

Components of PDA

- Finite Set of States (Q)
- Input Alphabet (Σ)
- Stack Alphabet (Γ)
- Transition Function (δ)
- Start State (q_0)
- Initial Stack Symbol (Z_0)
- Final States (F)

Stack Operations

Push: Insert a symbol onto the stack.

Pop: Remove a symbol from the stack.

Peek: Examine the top stack symbol without removing it.

Instantaneous Description (ID)

An Instantaneous Description represents the current configuration of a PDA including the current state, remaining input, and stack contents.

Acceptance by Final State

A string is accepted if the PDA reaches a final state after processing the entire input.

Acceptance by Empty Stack

A string is accepted if the PDA empties its stack after processing the entire input.

PDA Construction for $a^n b^n$

Step 1: Push one symbol for each 'a'.

Step 2: Pop one symbol for each 'b'.

Step 3: Accept when stack becomes empty after all input is processed.

CFG to PDA Conversion

Every Context-Free Grammar can be converted into an equivalent PDA by simulating grammar productions using stack operations.

Example

Grammar:

$S \rightarrow aSb \mid ab$

PDA pushes symbols corresponding to grammar variables and replaces them according to productions while matching input symbols.

Applications of PDA

- Compiler Design
- Syntax Analysis
- Parsing Algorithms
- Expression Evaluation
- Natural Language Processing

Exam Important Questions

1. Define Pushdown Automata.
2. Explain the components of PDA.
3. Describe stack operations.
4. Differentiate acceptance by final state and empty stack.
5. Construct a PDA for $a^n b^n$.
6. Explain CFG to PDA conversion.

Summary

Pushdown Automata are more powerful than finite automata because they use a stack for memory. They are fundamental for recognizing context-free languages and understanding parsing techniques.